TEXORA · LECTURE SHEET

Maintenance

Maintenance is the economics of failure: every machine will fail — the only choice is whether it fails at a planned moment, cheaply, or in mid-production, expensively. This sheet classifies maintenance types, contrasts scheduled with breakdown maintenance, and shows how maintenance is planned and organised in a textile mill, with the availability calculation that measures success.

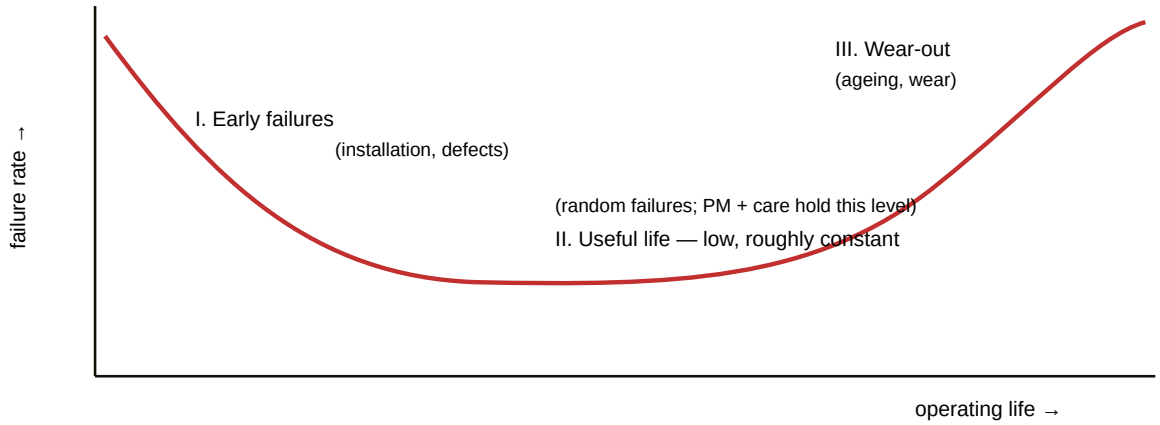
TE-4105 · INDUSTRIAL UTILITIES · TOPIC 9 · LEVEL 4 TERM 1

1. What maintenance is

Maintenance is all work that keeps plant and machinery in, or restores them to, a condition in which they can perform their required function at the required quality and rate: cleaning, lubrication (Topic 7), inspection, adjustment, repair and replacement. Its objectives are maximum **availability**, product quality, safety (Topic 11) and minimum total cost — where "total" includes the production lost while a machine stands, not merely the fitter's bill.

2. Types of maintenance

- **Breakdown (corrective / run-to-failure) maintenance:** repair only after failure. Rational only for cheap, non-critical items whose failure stops nothing and endangers nobody.
- **Preventive maintenance (PM):** planned inspection, servicing and part replacement at fixed intervals (time- or production-based) *before* failure — lubrication rounds, belt and filter changes, loom overhauls by schedule.
- **Predictive (condition-based) maintenance:** monitor the machine's actual condition — vibration, temperature, oil analysis (TAN, particles), thermography — and intervene when measurements show deterioration. Work is done when needed, neither early nor late.
- **Scheduled maintenance** is the umbrella term for work done to a timetable (PM and planned overhauls, §3); it contrasts with unscheduled breakdown work.
- **Improvement / design-out maintenance:** modify the machine or its setting to remove a recurring failure cause permanently.
- **Total productive maintenance (TPM):** an organisation philosophy in which operators own daily care (cleaning, oiling, tightening, inspection) and specialists own planned and predictive work — "zero breakdown" as a production goal.



The bathtub curve — failure rate against life. Good erection (Topic 8) shortens region I; preventive care lengthens region II.

Figure 1 — The bathtub curve. Failure rate is high in infancy, low and flat in useful life, and rises again at wear-out — the pattern that decides overhaul and replacement timing.

3. Scheduled maintenance

Scheduled maintenance is work whose content, timing and resources are decided in advance. A schedule is built from the maker's manual, failure history and production calendar:

- **Daily/weekly:** operator cleaning, blow-down of fly, oil-level checks, lubrication rounds, belt and temple checks on looms.
- **Monthly/quarterly:** inspections, filter changes, safety-device tests (boiler mountings, Topic 3), alignment and tension checks, dyeing-machine seal and valve servicing.
- **Annual/major:** overhauls — strip, measure wear against limits, replace worn parts (bearings, gears, healds/reeds by condition), re-level (Topic 8), repaint and return to service with a trial run.

Scheduled work is loaded into low-production windows (holidays, style changes) so its downtime costs least.

4. Breakdown maintenance

Breakdown maintenance is unplanned repair after a failure has stopped (or degraded) production. Its hidden costs dominate: lost output at the bottleneck, overtime, expedited spares, quality faults produced before the stop, restart waste, and safety risk from hurried work. A mill run mainly on breakdown maintenance is permanently firefighting: the workshop is full, yet availability is low. The policy goal is not zero breakdown work — random failures exist — but a small, shrinking share of it.

ASPECT	SCHEDULED (PREVENTIVE)	BREAKDOWN (CORRECTIVE)
Timing	Chosen — low-cost windows	Imposed by the failure

Downtime	Short, prepared, parts ready	Long — diagnosis + waiting for parts
Cost pattern	Steady, budgeted	Spiky; production loss dominates
Quality effect	Machine kept in tolerance	Faults produced before failure noticed
Safety	Planned isolation, permits	Rushed work, higher risk
Best for	Critical, wearing, quality-deciding plant	Cheap, non-critical, failure-proof items only

5. Measuring performance — availability

$$\text{Availability } A = \text{MTBF} \div (\text{MTBF} + \text{MTTR})$$

MTBF (mean time between failures) measures reliability; **MTTR** (mean time to repair) measures maintainability — diagnosis speed, spares readiness and fitter skill.

WORKED EXAMPLE — AVAILABILITY

A loom section records MTBF = 400 h and MTTR = 8 h.

A = 400 / (400 + 8) = 400/408 = 0.980 ≈ **98.0%**.

Interpretation: in every 408 h cycle the section is down for repair for 8 h. Note the leverage: halving MTTR to 4 h (spares kitted, trained crew) gives 400/404 ≈ 99.0% — the same gain as doubling MTBF to 800 h at 8 h MTTR (800/808 ≈ 99.0%), usually at far lower cost. Availability work therefore attacks repair time as hard as failure frequency.

6. Planning and organising maintenance

Planning (what, how, with what, how long):

1. **Asset register:** every machine numbered, with its file — manuals, erection records (Topic 8), lubrication chart (Topic 7), history card.
2. **Work definition:** PM task lists and intervals per machine; standard job sheets for overhauls.
3. **Spares planning:** classify spares by criticality; stock insurance spares for bottleneck machines; kit parts before a scheduled job.
4. **Scheduling:** weekly plan agreed with production; shutdown plans for major work; permits and isolation for safety.
5. **Recording:** every job logged — failure cause codes feed the analysis that improves the plan (a simple CMMS or even disciplined registers).

Organisation (who): a maintenance manager/engineer owning the plan; shift fitters for first response; specialist gangs (electrical, mechanical, instrumentation); operator-care duties under TPM; a central workshop and stores. Key indicators reviewed monthly: availability, breakdown share of hours, PM compliance (% of scheduled jobs done on time), repeat-failure list, maintenance cost per unit of production.

Glossary

Preventive maintenance — planned servicing before failure. **Breakdown maintenance** — repair after failure. **Predictive maintenance** — intervention triggered by measured condition. **MTBF** — mean time between failures. **MTTR** — mean time to repair. **Availability** — MTBF/(MTBF + MTTR); the share of time plant is able to run. **TPM** — total productive maintenance, operator-owned daily care plus specialist planned work. **CMMS** — computerised maintenance management system.

Key points

• Types: breakdown, preventive, predictive, improvement — scheduled work is planned work of any of these kinds.
• Bathtub curve: infant failures → flat useful life → wear-out; PM stretches the flat region.
• Availability example: 400/(400 + 8) ≈ 98.0%; cutting MTTR is often the cheapest availability gain.
• Planning = register, task lists, spares, schedule, records. Organisation = clear ownership plus monthly indicators.

Practice

1. A machine has MTBF = 250 h and MTTR = 10 h. Calculate availability. What MTTR would give 99% at the same MTBF?
2. Classify each job as breakdown, preventive or predictive: (a) replacing a pump after it seizes mid-shift; (b) quarterly bearing replacement by hours run; (c) bearing replacement after vibration analysis shows a defect growing.
3. Give three reasons breakdown maintenance costs more than its repair bill.
4. Name the five planning elements of §6 and state what each prevents.

Answer key

1. A = 250/260 ≈ **96.2%**. For 99%: 0.99 = 250/(250 + MTTR) → 250 + MTTR = 252.5 → MTTR ≈ **2.5 h**.
2. (a) Breakdown. (b) Preventive (time-based). (c) Predictive (condition-based).

3. Any three: lost production at possibly the bottleneck; defective output before the stop; overtime and expedited spares; restart/scrap waste; safety exposure of hurried repair; damage spreading to mating parts.
4. Asset register (prevents unknown plant/no history); work definition (prevents vague, variable jobs); spares planning (prevents waiting for parts); scheduling with production (prevents downtime at peak cost); recording and analysis (prevents repeat failures staying invisible).
